

ACOL216: Tutorial 8

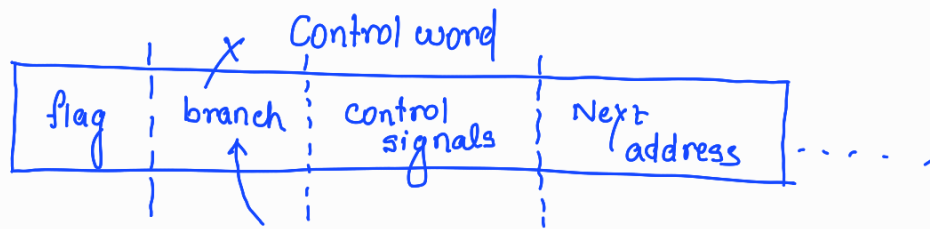
Date: April 6, 2026

1. Consider a single address field format used for branch control logic. Assume that the control word is 64 bits wide. The control portion of the micro-instruction format is divided into two fields. A micro-operation field of 24 bits specifies the micro-operations to be performed. An address selection field specifies a condition, based on the flags, that will cause a micro-instruction branch. There are 16 flags. How many bits are there in address bits? If it has a byte oriented memory then what is the size of the memory?
2. Consider a CPU with 128 instructions in the instruction sets, and each operation requires 8 cycles to finish. If 60 control signals are needed to be generated control units and if signal address field format is used, What is the minimum size of control word in the Horizontal microprogrammed control unit?
3. Consider a hypothetical control unit, which supports 512 bytes of control word memory. If the control word is 20 bits long and hardware contains 4 flags and supports 16 branch conditions, what is the maximum number of control signals that can be generated at a time using horizontal micro-programming and vertical micro-programming respectively?
4. Consider a microprogrammed control unit supports 7 different mutual exclusive groups of control signal as given below:

Control Groups	Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7
Control Signals	2	10	4	1	18	23	6

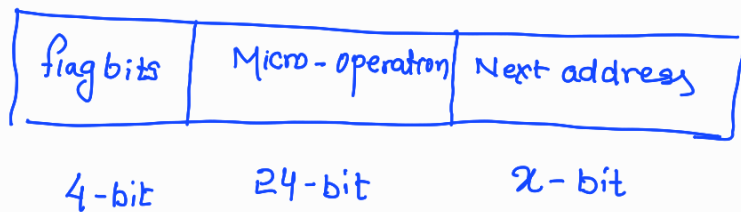
Now what is the minimum number of control bits required in both horizontal and vertical microprogrammings?

① Control word = 64-bit



There are 24-bits for Micro-Operations
and 16 flags $\rightarrow$ 4-bits

Control word = 64-bit



$$\Rightarrow 4 + 24 + x = 64$$

$$\Rightarrow x = 64 - 28 = 36\text{-bits}$$

If you are given with a byte oriented memory,

what is the size?

$$2^{36} \text{ B} \rightarrow 2^6 \times \underbrace{2^{30}}_{2^6} \text{ B} \rightarrow 64 \text{ GB}$$

Memory Size = 64GB

Control word = 64-bit
8B

②

Instruction set, 128 instructions

Each instruction will take 8 clock cycles to complete

That means 8 micro-operations required to complete 1 instruction,

Total number of Micro-instructions = 128×8

Bits required to store micro-instructions

$$= \log_2(128 \times 8) = 10\text{-bits}$$

There are 60 control signals,

Now for a Horizontal M.C.U = $60 + 10 = 70\text{-bit}$

Control word size = 70-bits

Now for a Vertical M.C.U = $\lceil \log_2 60 \rceil + 10 = 16\text{-bit}$

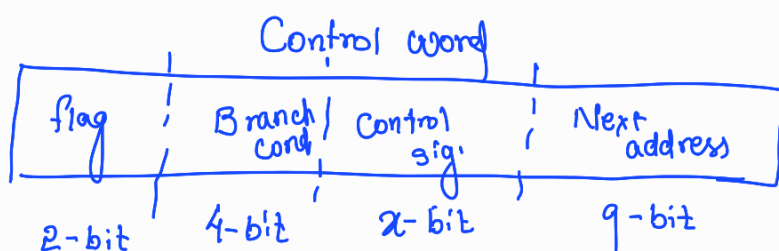
Control word size = 16-bits

③

512 B $\longrightarrow$ Control memory $\longrightarrow$ 9-bit

4 flags $\longrightarrow$ 2-bits

16 Branch Cond. $\longrightarrow$ 4-bits



$$\Rightarrow 2 + 4 + x + 9 = 20$$

$$\Rightarrow x = \underline{\underline{5\text{-bit}}}$$

Horizontal M.P.C.U = 5 control signals

Vertical M.P.C.U = $2^5 = 32$ control signals

At a time you can activate = 5 C.S. in the case of H.M.P.C.U.

But at a time you can only activate 1 C.S in the case of V.M.P.C.U.

(4)

G1 $\rightarrow 2 \rightarrow 1\text{-bit}$

G2 $\rightarrow 10 \rightarrow 4\text{-bit}$

G3 $\rightarrow 4 \rightarrow 2\text{-bit}$

G4 $\rightarrow 1 \rightarrow 1\text{-bit}$

G5 $\rightarrow 18 \rightarrow 5\text{-bit}$

G6 $\rightarrow 23 \rightarrow 5\text{-bit}$

G7 $\rightarrow 6 \rightarrow 3\text{-bit}$

H.M.P.C.U = $(2 + 10 + 4 + 1 + 18 + 23 + 6)$ bits

V.M.P.C.U = Since each group is independent and Mutually Exclusive,

You need to go each group and find the bits to represent,

$\rightarrow (1 + 4 + 2 + 1 + 5 + 5 + 3)$ bits